About This Agent

This AI agent demonstrates **Agentic AI** principles:

🧠 **Intelligence**: Uses Claude 4.5 Sonnet to understand locations and plan API calls

🔗 **Action**: Automatically calls the National Weather Service API

📊 **Processing**: Converts complex weather data into readable forecasts

💬 **Response**: Provides helpful, practical weather information

🏗️ Architecture

1. **User Input** → Location name
2. **AI Planning** → Generate API calls
3. **Points API** → Get forecast office
4. **Forecast API** → Get weather data
5. **AI Processing** → Create summary
6. **Display Results** → Show to user

🌤️ Weather AI Agent

Powered by Claude 4.5 Sonnet on Amazon Bedrock

This intelligent agent helps you get weather forecasts using the National Weather Service API. Enter any location below and watch the AI agent work through its reasoning process!

🔍 Enter a location name or description:

Times Square, NYC • Grand Canyon • 94129

🚀 Get Weather Forecast

🗑️ Clear Results

Weather Analysis for: Times Square, NYC • Grand Canyon • 94129

🧠 **Step 1: AI Planning Phase**

✅ Points API URL generated successfully!

```
https://api.weather.gov/points/40.758,-73.9855
```

🔗 **Step 2: Points API Execution**

✅ Received location data from NWS

🔍 View Raw Points API Response (first 500 characters)

📍 **Step 3: Extracting Forecast URL**

✅ Forecast URL extracted successfully!

```
https://api.weather.gov/gridpoints/OKX/34,37/forecast
```

🌦️ **Step 4: Forecast API Execution**

✅ Received 13,464 characters of forecast data

🔍 View Raw Forecast API Response (first 500 characters)

📊 **Step 5: AI Analysis Phase**

✅ Weather analysis complete!

📊 Process Status

Agent Workflow:
- ✅ Planning Phase
- ✅ Points API Call
- ✅ URL Extraction
- ✅ Forecast API Call
- ✅ Data Processing
- ✅ Results Generated

🎯 What Makes This Agentic?

- 🧠 **Reasoning**: AI understands location formats
- 📋 **Planning**: Generates appropriate API call sequences
- 🔧 **Action**: Executes real-world API requests
- 📊 **Processing**: Converts raw data to insights
- 🔄 **Adaptation**: Handles different location types

🌤️ Weather Forecast

Weather Forecast Summary

Location: Times Square, New York City *Forecast issued: February 28, 2026 at 2:10 PM EST*

Current Conditions & Today

This Afternoon (Saturday): Pleasant late-winter weather with mostly sunny skies. Temperatures will reach 51°F before cooling to around 49°F by evening. Light south winds at 5-8 mph. No precipitation expected.

Tonight: Mostly cloudy with areas of fog developing between 11 PM and 1 AM. Temperature dropping to 31°F, then rising to around 33°F overnight. Light southeast winds at 3-7 mph.

Next 3 Days Outlook

Sunday, March 1st ❄️

- **High:** 37°F
- **Conditions:** Light snow likely before 4 PM
- **Precipitation Chance:** 60%
- **Snow Accumulation:** Less than half an inch possible
- **Wind Chill:** As low as 21°F
- **Wind:** North at 7-10 mph
- **Night Low:** 17°F (feels like 8°F) - mostly clear skies

Monday, March 2nd ☀️

- **High:** 31°F | **Low:** 22°F
- **Conditions:** Sunny and cold during the day, partly cloudy at night
- **Precipitation:** None expected
- **Wind:** West at 7 mph, becoming south at 5 mph overnight

Tuesday, March 3rd 🌨️ 🌧️

- **High:** 41°F | **Night Low:** 38°F
- **Conditions:** Rain and snow likely after 7 AM, transitioning to rain
- **Precipitation Chance:** 70% (day), 91% (night)
- **Snow Accumulation:** Less than half an inch possible
- **Wind:** South at 3-9 mph, becoming southwest at 10 mph

Notable Weather Patterns

⚠️ **Temperature Roller Coaster:** Expect significant temperature swings over the next week, from a frigid 17°F Sunday night to milder 50s by mid-week.

❄️ **Winter Weather Alert:** Two snow events expected (Sunday and Tuesday morning), though accumulations should remain light (under half an inch each time).

🌧️ **Wet Mid-Week:** Tuesday night through Friday will be predominantly wet with multiple rounds of rain. Wednesday shows the highest precipitation chance at 80%.

💨 **Wind Chill Concerns:** Sunday will feel much colder than actual temperatures, with wind chills dropping to 8-21°F.

Planning Recommendations

- **This Weekend:** Bundle up for Sunday's snow and bitter cold Sunday night
- **Early Next Week:** Monday offers a brief dry window, though still quite cold
- **Mid-Week:** Keep umbrellas handy Tuesday through Friday with frequent rain expected
- **Temperature Trend:** Gradual warming from the 30s to low-50s by week's end

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🔬 About This Demo

This application demonstrates **Agentic AI** principles using:

- **Amazon Bedrock** with Claude 4.5 Sonnet for intelligent reasoning
- **National Weather Service API** for real-time weather data
- **Streamlit** for interactive web interface

⚠️ **Important**: This uses official NWS data for educational purposes. For critical weather decisions, consult official sources.

💡 Try These Examples:

Suggested locations to test:

- **Seattle** - Major city (tests city name recognition)
- **90210** - ZIP code (tests postal code handling)
- **New York City** - Multi-word city (tests complex location parsing)
- **Miami, FL** - City with state (tests state abbreviations)
- **Chicago** - Another major city (tests different coordinates)
- **National park near Homestead in Florida** - Location description (tests AI reasoning)
- **Largest City in California** - Descriptive query (tests knowledge-based location finding)

Simply copy any of these into the location input above and click "Get Weather Forecast"!